Watermarks for Language Models

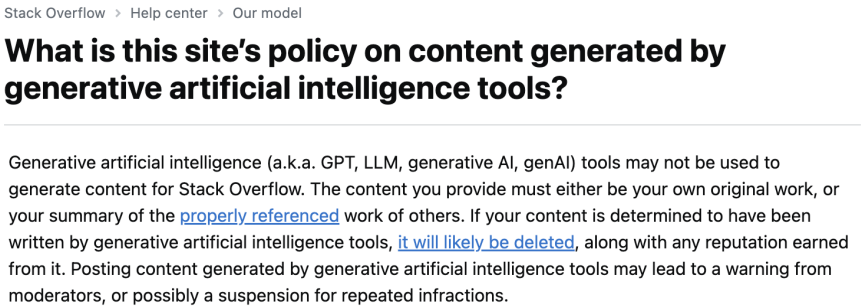


Seminar in AI - 15.05.2025
Tim Michael Toller, Supervisor: A.Univ.-Prof., DI Dr. Josef Küng
Institute for Application-oriented Knowledge Processing

Introduction & Motivation



Why watermark a language model?



Stack Overflow › Help center › Our model

What is this site's policy on content generated by generative artificial intelligence tools?

Generative artificial intelligence (a.k.a. GPT, LLM, generative AI, genAI) tools may not be used to generate content for Stack Overflow. The content you provide must either be your own original work, or your summary of the properly referenced work of others. If your content is determined to have been written by generative artificial intelligence tools, it will likely be deleted, along with any reputation earned from it. Posting content generated by generative artificial intelligence tools may lead to a warning from moderators, or possibly a suspension for repeated infractions.

Figure: Screenshot of the Stack Overflow generative AI policy[2].

Why watermark a language model?

- Large language models can mass-produce human-like text, raising concerns about *provenance*[1].
- The increased usage of LMs can lead to issues in academic integrity, plagiarism, copyright infringement and the propagation of misinformation
- Platforms and developers of LMs want to track (mis)use of their models.
- Can improve future LM training, as text generated by LMs can be removed from the training set.

[1] a record of ownership of a work of art or an antique, used as a guide to authenticity or quality

What is watermarking?

- A **watermark** is a hidden signal embedded in generated text that encodes its source. This signal can be detected later to prove the text's origin.
- The watermarking protocol described by *Kuditipudi et al.* involves three parties:
  1. **LM Provider:** Generates text with a secret watermark (using a secret key).
  2. **User:** Receives the watermarked text. They might publish it or attempt to *edit it to evade detection*.
  3. **Detector:** Uses the shared secret key to verify if given text carries the watermark (hence came from the provider).

Note

In this scenario, the **LM Provider** and the **Detector** work together to generate and verify the watermark. The **User** is considered an adversary trying to remove the watermark. [1]

Example Flow

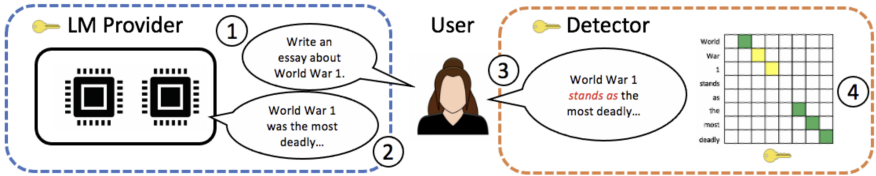


Figure: Example Watermarking protocol.[1]

1. The **User** requests a text from the **LM Provider**.
2. The **LM Provider** generates the text and embeds a watermark using a secret key.
3. The (malicious) **User** receives the watermarked text, edits it, and publishes it.
4. The **Detector** receives the text and uses the secret key to check if it contains a watermark. Even if the text is edited, the watermark should still be detectable.

Properties of an ideal Text Watermark

1. **Distortion-free:** Preserves the model's *original output distribution*. The watermark shouldn't degrade quality or bias the content.
2. **Agnostic detection:** Can be detected *without access to the original model or prompt*. There should be need to query the model or know the context.
3. **Robust:** Should *withstand perturbations* of the watermarked text such as editing, paraphrasing, cropping, or noise insertion in the text.

Hands-on Example



Hands-on example

We will now walk through a simple, yet full hands-on example of a distortion-free watermarking protocol based on [1].

Parameter	Value
model	huggyllama/llama-7b
prompt	“Steganography is"
m (generated text length)	100
n (watermark sequence length)	256
key	42
seed	1
method	Exp-Min + Levenshtein

Table: Parameters for our watermarking example [3]

Step 1: Generating the watermark

We first generate a response based on the prompt with no watermarking. This will act as a control text. We also generate a watermarked text using the same prompt and the same model.

Plain/Control text (p=0.8020)	**Watermarked (p=0.0099)**
Steganography is a technique used to hide secret messages within an image, a song, or other media that are not usually used to convey secure information. Steganography can be used to conceal the fact that there is a secret message in the first place, but it can also be used to transmit secret messages. There are many different methods of encoding secret messages in an image...	Steganography is the practice of hiding information or data inside another file or software even if a malicious party tries their hardest to reveal the hidden codes. It basically closes the door to hackers and only allotts access to those who are suppose to take it. This makes it, in a sense, lie to the user and or viewer of the data and anyone trying to open the file...

Step 2: Editing the text

We want to ensure the watermark is robust against rewriting or paraphrasing. To test this, we translate the watermarked text into different languages and back ($EN \rightarrow FR \rightarrow EN \rightarrow RU \rightarrow EN$). This results in about 25% of the text being altered with the content staying mostly the same.

Watermarked (p=0.0099)	**Edited (p=0.0099)**
Steganography is the practice of hiding information or data ~~inside~~ another file or software even if ~~a malicious party tries their hardest to reveal the hidden codes~~. It ~~basically~~ closes the ~~door to hackers~~ and only ~~allotts~~ access to those who ~~are suppose to take it~~. This ~~makes it,~~ in a sense, lie to the user ~~and or viewer of the data~~ and anyone ~~trying~~ to open the file...	Steganography is the practice of hiding information or data in another file or software, even if the malicious side tries to open hidden codes. In essence, it closes the pirate door and attributes only access to those who should take it. This is, in a sense, to lie to the user / or a data viewer and anyone who tries to open the file...

Step 3: Checking the watermark

To reiterate the p-values we obtained, we have:

Plain/Control	Watermarked	Edited
p=0.8020	p=0.0099	p=0.0099

A text is considered watermarked if **p < 0.01**.

As expected, the control text is not watermarked, while the watermarked and edited texts are both watermarked.

Methodology



Method comparisons

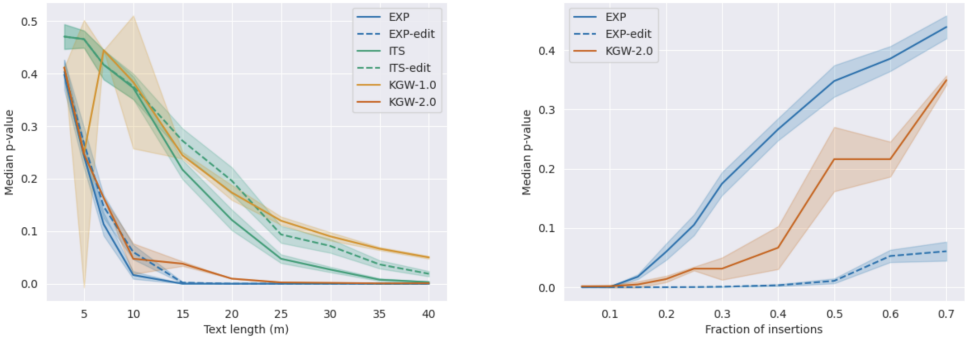


Figure: Comparing Exp-Min & ITS (with and without Levenshtein) and Kirchenbauer et al. [1].

Preferred Method

- Distortion-free decoding using **Exponential-Min** sampling. For a language-model next-token distribution $p : \mathcal{V} \to [0, 1]$ and a private key $\xi_i \sim \mathcal{U}(0, 1)^{|\mathcal{V}|}$, select

$$t_i^\star \;=\; \arg\min_{t \in \mathcal{V}} \left\{ -\frac{\ln \xi_{i,t}}{p(t)} \right\},$$

  which samples $t_i^\star \sim p$ without altering p. This method will be used for the following examples.
- Robustness through **Levenshtein cost**.
- With these methods, the watermark starts to be reliable at around **m > 35** generated tokens and stays robust even after corrupting **40%-50%** of the text.

Generation process

1. **Key generation.** Seed a Mersenne–Twister PRNG with the secret key and draw an n-element stream $(\xi_i \sim \mathcal{U}(0,1)^{|\mathcal{V}|})$, which is stored for later use.
2. **Random shift.** At generation time, obtain a random offset $\tau \sim \mathcal{U}\{0, n-1\}$ (independent of the key) and set

$$\texttt{shift}(\xi, \tau) = (\xi_\tau, \xi_{\tau+1}, \ldots, \xi_{\tau+m-1}),$$

   where m is the number of tokens to be generated.
3. **Token sampling.** At step i apply Exp-Min with the shifted key vector ξ_i to draw $t_i^\star$. High-probability LM tokens remain likely; ties between similarly probable tokens are biased toward *higher* ξ values, yielding the hidden correlation.
4. **Detection.** Given text **t**, compute the *Levenshtein-cost* and compare it with the same cost under random surrogate keys; a small p-value ($p < 0.01$) signals a watermark.

Token-selection example

Prompt: *Steganography is the practice of hiding*, $\tau = 41$ [3]

token	p_{LM}	ξ	$w(t) \propto \exp\left(-\frac{\ln \xi}{p_{\text{LM}}}\right)$
messages	0.2000	0.5555	0.1517
information	0.1943	0.7710	**0.7513**
data	0.1500	0.3428	0.0023
a	0.1323	0.2594	0.0001
secret	0.1252	0.1291	$\ll 10^{-4}$
the	0.0290	0.8610	0.0165
...	...	...	...

messages and *information* have similar LM probabilities, but the larger ξ for *information* gives it bias under the watermark. Unlikely tokens never become likely under the watermark, so sampling remains distortion-free.

Outlook



Outlook & Personal opinion

- The *Kuditipudi et al.* watermark described in this presentation works only if the **LM Provider** and the **detector** work together. This is unreliable once users move to open-source LMs or switch providers.
- Beyond scrubbing a watermark, we may want to *prove* human authorship or certify other attributes, each needing a different embed/detect design and careful control of false positives/negatives.
- There are many different ideas and methods to choose from, it's still not quite clear which approach is best long-term.
- Research should target provider-agnostic, robust schemes. Adoption could arrive through regulation or market incentives.
- Major LMs may already watermark their text and we wouldn't be able to tell.

References

[1] Rohith Kuditipudi, John Thickstun, Tatsunori Hashimoto, and Percy Liang. 2023. Robust Distortion-free Watermarks for Language Models. arXiv: 2307.15593 [cs.LG]. https://crfm.stanford.edu/2023/07/30/watermarking.html.

[2] Stack Overflow. 2022. Temporary policy: Generative AI on Stack Overflow. Retrieved 04/26/2025 from https://stackoverflow.com/help/gen-ai-policy.

[3] Tim Toller. 2025. Robust Distortion-free Watermarking for Language Models: Jupyter Notebook Implementation and Extensions. Jupyter Notebook, Appendix. Based on the code of Kuditipudi et al. (2023), https://github.com/jthickstun/watermark/, with additional modifications. (2025).



JOHANNES KEPLER
UNIVERSITY LINZ

Appendix



implementation_demo

May 6, 2025

1 LM Watermarking implementation Demo

This notebook is heavily based on the work here https://github.com/jthickstun/watermark/

The notebook was executed on a google colab project online.

[1]:
```
# Notebook parameters (modify as needed)#
#model_name = "facebook/opt-2.7b"
model_name = "huggyllama/llama-7b"
prompt = "Steganography is"
m = 100             # length of generation
n = 256            # watermark sequence length
key = 42           # watermark key
seed = 1           # random seed for reproducibility
```

[2]:
```
import torch
import numpy as np
import random
import tqdm
from transformers import AutoTokenizer, AutoModelForCausalLM

device = torch.device(
    'cuda' if torch.cuda.is_available() else
    ('mps' if torch.backends.mps.is_available() else 'cpu')
)
print(device)
```

```
cpu
```

[3]:
```
# Seed for reproducibility
torch.manual_seed(seed)
np.random.seed(seed)
random.seed(seed)

# Initialize tokenizer and model
tokenizer = AutoTokenizer.from_pretrained(model_name)

model = AutoModelForCausalLM.from_pretrained(
    model_name,
```

```
    device_map="auto",
)

# Prepare inputs
inputs = tokenizer.encode(prompt, return_tensors='pt').to(device)
```

```
/usr/local/lib/python3.11/dist-packages/huggingface_hub/utils/_auth.py:94:
UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab
(https://huggingface.co/settings/tokens), set it as secret in your Google Colab
and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access
public models or datasets.
  warnings.warn(
You are using the default legacy behaviour of the <class
'transformers.models.llama.tokenization_llama_fast.LlamaTokenizerFast'>. This is
expected, and simply means that the `legacy` (previous) behavior will be used so
nothing changes for you. If you want to use the new behaviour, set
`legacy=False`. This should only be set if you understand what it means, and
thoroughly read the reason why this was added as explained in
https://github.com/huggingface/transformers/pull/24565 - if you loaded a llama
tokenizer from a GGUF file you can ignore this message.

Loading checkpoint shards:   0%|          | 0/2 [00:00<?, ?it/s]
```

[4]:
```
# Generate normally
outputs = model.generate(
    inputs,
    max_new_tokens=m,
    do_sample=True,
)

# Decode and display
normal_text = tokenizer.decode(outputs[0], skip_special_tokens=True)

print("--- Normal Output ---")
print(normal_text)
```

```
--- Normal Output ---
Steganography is a technique used to hide secret messages within an image, a
song, or other media that are not usually used to convey secure information.
Steganography can be used to conceal the fact that there is a secret message in
the first place, but it can also be used to transmit secret messages. There are
many different methods of encoding secret messages in an image. Some of the
methods are:
Compression method (also called Spread spectrum)
The key challenge with steganography is embedding
```

[5]:
```python
# Mersenne Twister based RNG for watermarking
def int_32(x): return int(0xFFFFFFFF & x)

class mersenne_rng:
    def __init__(self, seed=5489):
        self.state = [0]*624
        self.index = 624
        self.upper_mask = (1<<31)
        self.lower_mask = (1<<31) - 1
        # constants
        self.f, self.m = 1812433253, 397
        self.u, self.s, self.b = 11, 7, 0x9D2C5680
        self.t, self.c, self.l = 15, 0xEFC60000, 18
        # initialize state
        self.state[0] = seed
        for i in range(1, 624):
            prev = self.state[i-1]
            self.state[i] = int_32(self.f * (prev ^ (prev >> 30)) + i)

    def twist(self):
        for i in range(624):
            x = (self.state[i] & self.upper_mask) + (self.state[(i+1)%624] &␣
 ↪self.lower_mask)
            xA = x >> 1
            if x % 2: xA ^= 0x9908b0df
            self.state[i] = self.state[(i + self.m) % 624] ^ xA
        self.index = 0

    def rand(self):
        if self.index >= 624:
            self.twist()
        y = self.state[self.index]
        y ^= (y >> self.u)
        y ^= ((y << self.s) & self.b)
        y ^= ((y << self.t) & self.c)
        y ^= (y >> self.l)
        self.index += 1
        return (int_32(y) * (1.0 / 4294967296.0))
```

[6]:
```python
# Exponential watermark sampling
def exp_sampling(probs, u):
    # probs: tensor [vocab], u: numpy array [vocab]
    return torch.argmax(torch.from_numpy(u) ** (1.0 / probs.cpu().numpy()),␣
 ↪dim=1).unsqueeze(-1)

def generate_shift(model, prompt_ids, vocab_size, n, m, key):
    rng = mersenne_rng(key)
```

```
    # create watermark matrix xi
    xi = np.array([rng.rand() for _ in range(n * vocab_size)], dtype=np.
↪float32).reshape(n, vocab_size)
    torch.manual_seed(seed)
    shift = torch.randint(0, n, (1,)).item()
    print(f"Shift: {shift}")

    inputs = prompt_ids.to(model.device)
    attn = torch.ones_like(inputs)
    past = None
    for i in tqdm.tqdm(range(m)):
        with torch.no_grad():
            if past:
                out = model(inputs[:, -1:], past_key_values=past,␣
↪attention_mask=attn)
            else:
                out = model(inputs)
        probs = torch.nn.functional.softmax(out.logits[:, -1, :vocab_size],␣
↪dim=-1)
        u = xi[(shift + i) % n]
        token = exp_sampling(probs, u)
        inputs = torch.cat([inputs, token.to(model.device)], dim=-1)
        past = out.past_key_values
        attn = torch.cat([attn, torch.ones((attn.shape[0],1), device=attn.
↪device)], dim=-1)

    return inputs.cpu()
```

[7]:
```
# Encode prompt & run watermark generation
tokens = tokenizer.encode(prompt, return_tensors='pt')
wm_tokens = generate_shift(model, tokens, len(tokenizer), n, m, key)[0]
watermarked_text = tokenizer.decode(wm_tokens, skip_special_tokens=True)
print()
print("--- Watermarked Output ---")
print(watermarked_text)
```

```
Shift: 37

  0%|          | 0/100 [00:00<?, ?it/s]<ipython-input-6-090363e8b445>:4:
DeprecationWarning: __array_wrap__ must accept context and return_scalar
arguments (positionally) in the future. (Deprecated NumPy 2.0)
  return torch.argmax(torch.from_numpy(u) ** (1.0 / probs.cpu().numpy()),
dim=1).unsqueeze(-1)
100%|     | 100/100 [01:18<00:00,  1.28it/s]


--- Watermarked Output ---
Steganography is the practice of hiding information or data inside another file
```

or software even if a malicious party tries their hardest to reveal the hidden codes. It basically closes the door to hackers and only allotts access to those who are suppose to take it. This makes it, in a sense, lie to the user and or viewer of the data and anyone trying to open the file.
Stegano is a useful tool for exchanging messages without anyone but the sender and receiver

[8]:
```
from numba import njit, prange
import math

@njit
def levenshtein(x, y, gamma=0.0):
    n = len(x)
    m = len(y)
    A = np.zeros((n + 1, m + 1), dtype=np.float64)
    for i in range(n + 1):
        for j in range(m + 1):
            if i == 0:
                A[i, j] = j * gamma
            elif j == 0:
                A[i, j] = i * gamma
            else:
                cost_sub = math.log(1.0 - y[j - 1, x[i - 1]])
                # deletion
                best = A[i - 1, j] + gamma
                # insertion
                tmp = A[i, j - 1] + gamma
                if tmp < best:
                    best = tmp
                # substitution
                tmp = A[i - 1, j - 1] + cost_sub
                if tmp < best:
                    best = tmp
                A[i, j] = best
    return A[n, m]
```

[9]:
```
import time

from numba import njit, prange


@njit
def detect(tokens, n, k, xi, gamma=0.0):
    m = len(tokens)
    n = len(xi)
```

```
    A = np.empty((m - (k - 1), n))
    for i in range(m - (k - 1)):
        for j in range(n):
            A[i][j] = levenshtein(tokens[i : i + k], xi[(j + np.arange(k)) %␣
 ↪n], gamma)

    return np.min(A)

def permutation_test(tokens, key, n, k, vocab_size, n_runs=100):
    rng = mersenne_rng(key)
    xi = np.array(
        [rng.rand() for _ in range(n * vocab_size)], dtype=np.float32
    ).reshape(n, vocab_size)
    test_result = detect(tokens, n, k, xi)

    p_val = 0
    for run in tqdm.tqdm(range(n_runs)):
        xi_alternative = np.random.rand(n, vocab_size).astype(np.float32)
        null_result = detect(tokens, n, k, xi_alternative)

        # assuming lower test values indicate presence of watermark
        p_val += null_result <= test_result

    return (p_val + 1.0) / (n_runs + 1.0)


# Run decoding test
tokenizer = AutoTokenizer.from_pretrained(model_name)
tokens = tokenizer.encode(watermarked_text, return_tensors='pt',␣
 ↪truncation=True, max_length=2048).numpy()[0]
tok_list = tokens.tolist()
print(tok_list)
start = time.time()
p_val_watermarked = permutation_test(tok_list, key, n, len(tok_list),␣
 ↪len(tokenizer))
print()
print(f"p-value: {p_val_watermarked:.4f} (elapsed {time.time()-start:.1f}s)")
print(f"Is watermarked: {p_val_watermarked<0.01}")
```

```
[1, 2443, 6249, 5275, 338, 278, 6944, 310, 25508, 2472, 470, 848, 2768, 1790,
934, 470, 7047, 1584, 565, 263, 4439, 14803, 6263, 14335, 1009, 2898, 342, 304,
10320, 284, 278, 7934, 11561, 29889, 739, 8830, 4694, 267, 278, 3050, 304,
15833, 414, 322, 871, 599, 327, 1372, 2130, 304, 1906, 1058, 526, 7755, 304,
2125, 372, 29889, 910, 3732, 372, 29892, 297, 263, 4060, 29892, 3804, 304, 278,
1404, 322, 470, 6316, 556, 310, 278, 848, 322, 5019, 1811, 304, 1722, 278, 934,
29889, 13, 7789, 29887, 1562, 338, 263, 5407, 5780, 363, 429, 305, 9776, 7191,
1728, 5019, 541, 278, 10004, 322, 19870]
```

```
100%|      | 100/100 [01:24<00:00,  1.18it/s]

p-value: 0.0099 (elapsed 99.3s)
Is watermarked: True
```

[10]:
```
!pip install googletrans==4.0.0-rc1
from googletrans import Translator

def back_translate(text, target_language='es'):
    """Translates text to a target language and back to the original language.
    """
    translator = Translator()
    translated = translator.translate(text, dest=target_language).text
    back_translated = translator.translate(translated, dest='en').text
    return back_translated

#def add_noise(text, noise_level=0.1):
#    """Adds random noise to the text."""
#    words = text.split()
#    num_noise_words = int(len(words) * noise_level)
#    noise_words = ["noise" for _ in range(num_noise_words)]
#    for i in range(num_noise_words):
#        words.insert(random.randint(0, len(words)), noise_words[i])
#    return " ".join(words)

# Apply back-translation to the watermarked text EN -> FR -> EN -> RU -> EN
back_translated_text = back_translate(back_translate(watermarked_text, "fr"),␣
    ↪"ru")
print("Back translated: ", back_translated_text)

# Encode the back-translated text
back_translated_tokens = tokenizer.encode(back_translated_text,␣
    ↪return_tensors='pt', truncation=True, max_length=2048).numpy()[0]
back_translated_tok_list = back_translated_tokens.tolist()

# Detect watermark in the back-translated text
start = time.time()
p_val_back_translated = permutation_test(back_translated_tok_list, key, n,␣
    ↪len(back_translated_tok_list), len(tokenizer))
print()
print(f"p-value (back-translated): {p_val_back_translated:.4f} (elapsed {time.
    ↪time()-start:.1f}s)")
print(f"Is watermarked (back-translated): {p_val_back_translated<0.01}")
```

```
Requirement already satisfied: googletrans==4.0.0-rc1 in
/usr/local/lib/python3.11/dist-packages (4.0.0rc1)
```

```
Requirement already satisfied: httpx==0.13.3 in /usr/local/lib/python3.11/dist-
packages (from googletrans==4.0.0-rc1) (0.13.3)
Requirement already satisfied: certifi in /usr/local/lib/python3.11/dist-
packages (from httpx==0.13.3->googletrans==4.0.0-rc1) (2025.4.26)
Requirement already satisfied: hstspreload in /usr/local/lib/python3.11/dist-
packages (from httpx==0.13.3->googletrans==4.0.0-rc1) (2025.1.1)
Requirement already satisfied: sniffio in /usr/local/lib/python3.11/dist-
packages (from httpx==0.13.3->googletrans==4.0.0-rc1) (1.3.1)
Requirement already satisfied: chardet==3.* in /usr/local/lib/python3.11/dist-
packages (from httpx==0.13.3->googletrans==4.0.0-rc1) (3.0.4)
Requirement already satisfied: idna==2.* in /usr/local/lib/python3.11/dist-
packages (from httpx==0.13.3->googletrans==4.0.0-rc1) (2.10)
Requirement already satisfied: rfc3986<2,>=1.3 in
/usr/local/lib/python3.11/dist-packages (from
httpx==0.13.3->googletrans==4.0.0-rc1) (1.5.0)
Requirement already satisfied: httpcore==0.9.* in
/usr/local/lib/python3.11/dist-packages (from
httpx==0.13.3->googletrans==4.0.0-rc1) (0.9.1)
Requirement already satisfied: h11<0.10,>=0.8 in /usr/local/lib/python3.11/dist-
packages (from httpcore==0.9.*->httpx==0.13.3->googletrans==4.0.0-rc1) (0.9.0)
Requirement already satisfied: h2==3.* in /usr/local/lib/python3.11/dist-
packages (from httpcore==0.9.*->httpx==0.13.3->googletrans==4.0.0-rc1) (3.2.0)
Requirement already satisfied: hyperframe<6,>=5.2.0 in
/usr/local/lib/python3.11/dist-packages (from
h2==3.*->httpcore==0.9.*->httpx==0.13.3->googletrans==4.0.0-rc1) (5.2.0)
Requirement already satisfied: hpack<4,>=3.0 in /usr/local/lib/python3.11/dist-
packages (from h2==3.*->httpcore==0.9.*->httpx==0.13.3->googletrans==4.0.0-rc1)
(3.0.0)
Back translated:  Steganography is the practice of hiding information or data in
another file or software, even if the malicious side tries to open hidden codes.
In essence, it closes the pirate door and attributes only access to those who
should take it. This is, in a sense, to lie to the user and / or a data viewer
and anyone who tries to open the file.
Stegano - a useful tool for exchanging messages without anyone, except the
sender and recipient

100%|      | 100/100 [01:22<00:00,  1.21it/s]


p-value (back-translated): 0.0099 (elapsed 94.7s)
Is watermarked (back-translated): True
```

[11]:
```
tokens_control = tokenizer.encode(normal_text, return_tensors='pt',␣
 ↪truncation=True, max_length=2048).numpy()[0]
tok_control_list = tokens_control.tolist()
print(tok_control_list)
start = time.time()
```

```
p_val_control = permutation_test(tok_control_list, key, n,␣
  ↪len(tok_control_list), len(tokenizer))
print()
print(f"p-value (control): {p_val_control:.4f} (elapsed {time.time()-start:.
  ↪1f}s)")
print(f"Is watermarked (control): {p_val_control<0.01}")
```

```
[1, 2443, 6249, 5275, 338, 263, 11043, 1304, 304, 9563, 7035, 7191, 2629, 385,
1967, 29892, 263, 4823, 29892, 470, 916, 5745, 393, 526, 451, 5491, 1304, 304,
27769, 11592, 2472, 29889, 2443, 6249, 5275, 508, 367, 1304, 304, 10628, 284,
278, 2114, 393, 727, 338, 263, 7035, 2643, 297, 278, 937, 2058, 29892, 541, 372,
508, 884, 367, 1304, 304, 22649, 7035, 7191, 29889, 1670, 526, 1784, 1422, 3519,
310, 8025, 7035, 7191, 297, 385, 1967, 29889, 3834, 310, 278, 3519, 526, 29901,
13, 1523, 2590, 1158, 313, 15189, 2000, 1706, 949, 18272, 29897, 13, 1576, 1820,
18766, 411, 1886, 6249, 5275, 338, 23655]
```

```
100%|      | 100/100 [01:35<00:00,  1.05it/s]
```

```
p-value (control): 0.8020 (elapsed 107.8s)
Is watermarked (control): False
```

[12]:
```
print(f"""
--- Plain/Control text (p={p_val_control:.4f}) ---
{normal_text}

--- Watermarked (p={p_val_watermarked:.4f}) ---
{watermarked_text}

--- Edited (p={p_val_back_translated:.4f}) ---
{back_translated_text}
""")
```

```
--- Plain/Control text (p=0.8020) ---
Steganography is a technique used to hide secret messages within an image, a
song, or other media that are not usually used to convey secure information.
Steganography can be used to conceal the fact that there is a secret message in
the first place, but it can also be used to transmit secret messages. There are
many different methods of encoding secret messages in an image. Some of the
methods are:
Compression method (also called Spread spectrum)
The key challenge with steganography is embedding

--- Watermarked (p=0.0099) ---
Steganography is the practice of hiding information or data inside another file
or software even if a malicious party tries their hardest to reveal the hidden
codes. It basically closes the door to hackers and only allotts access to those
```

```
who are suppose to take it. This makes it, in a sense, lie to the user and or
viewer of the data and anyone trying to open the file.
Stegano is a useful tool for exchanging messages without anyone but the sender
and receiver

--- Edited (p=0.0099) ---
Steganography is the practice of hiding information or data in another file or
software, even if the malicious side tries to open hidden codes. In essence, it
closes the pirate door and attributes only access to those who should take it.
This is, in a sense, to lie to the user and / or a data viewer and anyone who
tries to open the file.
Stegano - a useful tool for exchanging messages without anyone, except the
sender and recipient
```

1.1 Verbose Generation

Really go through the process step by step to see how each of the probabilites get affected.

[16]:
```python
def verbose_generate(model, prompt, n=256, m=20, key=42, top_k=10):
    """
    Generate *m* watermarked tokens from *prompt* while printing rich␣
 ↪diagnostics.
    """

    tokenizer  = AutoTokenizer.from_pretrained(model_name)
    vocab_size = len(tokenizer)

    rng = mersenne_rng(key)
    xi  = np.array([rng.rand() for _ in range(n * vocab_size)],
                   dtype=np.float32).reshape(n, vocab_size)
    torch.manual_seed(seed)
    shift = torch.randint(0, n, (1,)).item()
    print(f"Shift: {shift}")

    inputs = tokenizer.encode(prompt, return_tensors="pt").to(model.device)
    attn   = torch.ones_like(inputs)
    past   = None

    print(f"Initial prompt: {prompt!r}\n\n---  BEGIN VERBOSE GENERATION  ---")

    for i in range(m):
        with torch.no_grad():
            out = model(
                inputs if past is None else inputs[:, -1:],
                past_key_values=past,
                attention_mask=attn,
            )
```

```
            lm_probs = torch.softmax(out.logits[:, -1, :vocab_size], dim=-1)[0]

        j   = (shift + i) % n
        row = torch.from_numpy(xi[j]).to(lm_probs.device)

        wm_weights = torch.pow(row, 1.0 / lm_probs)
        wm_probs   = wm_weights / wm_weights.sum()

        top_idx = torch.topk(lm_probs, k=top_k).indices.tolist()

        print(f"\nStep {i:02d} - j = {j}")
        print(" token                          p_LM     xi      p_WM")
        print("                              ")
        for t in top_idx:
            token_str = tokenizer.decode([t]).replace("\n", "\\n")
            print(f"{token_str:<26}  {lm_probs[t]:.4f}  {row[t]:.4f}
↪{wm_probs[t]:.4f}")

        token = exp_sampling(lm_probs.unsqueeze(0), row.cpu().numpy()).item()
        chosen = tokenizer.decode([token]).replace("\n", "\\n")
        print(f" → Chosen token: {token:>4d}  ('{chosen}')")

        inputs = torch.cat([inputs, torch.tensor([[token]], device=inputs.
↪device)], dim=-1)
        past   = out.past_key_values
        attn   = torch.cat([attn, torch.ones((attn.shape[0], 1), device=attn.
↪device)], dim=-1)

        current_text = tokenizer.decode(inputs[0], skip_special_tokens=True)
        print(" Current prompt so far →", repr(current_text))

    print("---  END VERBOSE GENERATION  ---\n")
    return tokenizer.decode(inputs[0], skip_special_tokens=True)

verbose_text = verbose_generate(model, prompt, n=n, m=20, key=key)
print("\nFinal partial generation:\n", verbose_text)
```

```
Shift: 37
Initial prompt: 'Steganography is'

---  BEGIN VERBOSE GENERATION  ---

Step 00 - j = 37
 token                          p_LM     xi      p_WM

the                          0.6179  0.6642  0.9988
a                            0.1936  0.2224  0.0008
```

```
an                          0.0343  0.4728  0.0000
one                         0.0095  0.1159  0.0000
hiding                      0.0080  0.2565  0.0000
not                         0.0069  0.6270  0.0000
defined                     0.0057  0.4347  0.0000
used                        0.0051  0.1604  0.0000
basically                   0.0039  0.3203  0.0000
often                       0.0034  0.1888  0.0000
 → Chosen token:  278  ('the')
 Current prompt so far → 'Steganography is the'

<ipython-input-6-090363e8b445>:4: DeprecationWarning: __array_wrap__ must accept
context and return_scalar arguments (positionally) in the future. (Deprecated
NumPy 2.0)
  return torch.argmax(torch.from_numpy(u) ** (1.0 / probs.cpu().numpy()),
dim=1).unsqueeze(-1)


Step 01 - j = 38
 token                        p_LM     xi      p_WM

art                         0.4627  0.2408  0.0556
practice                    0.1638  0.9550  0.9110
process                     0.0665  0.1585  0.0000
science                     0.0655  0.7853  0.0302
technique                   0.0316  0.0601  0.0000
study                       0.0199  0.1586  0.0000
hiding                      0.0197  0.1910  0.0000
act                         0.0186  0.2022  0.0000
method                      0.0124  0.1854  0.0000
name                        0.0093  0.7448  0.0000
 → Chosen token: 6944  ('practice')
 Current prompt so far → 'Steganography is the practice'

Step 02 - j = 39
 token                        p_LM     xi      p_WM

of                          0.9547  0.4524  0.9978
and                         0.0235  0.3652  0.0000
or                          0.0077  0.2826  0.0000
,                           0.0040  0.7004  0.0000
that                        0.0019  0.9407  0.0000
to                          0.0012  0.2799  0.0000
by                          0.0008  0.0994  0.0000
in                          0.0007  0.8650  0.0000
used                        0.0006  0.1539  0.0000
where                       0.0005  0.2094  0.0000
 → Chosen token:  310  ('of')
 Current prompt so far → 'Steganography is the practice of'
```

```
Step 03 - j = 40
 token                          p_LM     xi      p_WM

hiding                         0.5877  0.2907  0.3969
conce                          0.2393  0.5335  0.2353
embedding                      0.0312  0.4600  0.0000
secret                         0.0145  0.0479  0.0000
writing                        0.0128  0.5202  0.0000
using                          0.0096  0.3912  0.0000
obsc                           0.0084  0.0113  0.0000
encoding                       0.0082  0.1361  0.0000
communic                       0.0081  0.6516  0.0000
encrypt                        0.0079  0.6208  0.0000
 → Chosen token: 25508  ('hiding')
 Current prompt so far → 'Steganography is the practice of hiding'

Step 04 - j = 41
 token                          p_LM     xi      p_WM

messages                       0.2000  0.5555  0.1517
information                    0.1943  0.7710  0.7513
data                           0.1500  0.3428  0.0023
a                              0.1323  0.2594  0.0001
secret                         0.1252  0.1291  0.0000
the                            0.0290  0.8610  0.0165
or                             0.0123  0.8681  0.0000
sensitive                      0.0106  0.2001  0.0000
something                      0.0105  0.7360  0.0000
text                           0.0075  0.8341  0.0000
 → Chosen token: 2472  ('information')
 Current prompt so far → 'Steganography is the practice of hiding information'

Step 05 - j = 42
 token                          p_LM     xi      p_WM

in                             0.4128  0.3204  0.0569
within                         0.2405  0.8240  0.4010
or                             0.0587  0.9707  0.5402
inside                         0.0473  0.3905  0.0000
,                              0.0425  0.3039  0.0000
.                              0.0272  0.1466  0.0000
(                              0.0262  0.1873  0.0000
by                             0.0155  0.3577  0.0000
and                            0.0139  0.4302  0.0000
to                             0.0098  0.4717  0.0000
 → Chosen token:  470  ('or')
 Current prompt so far → 'Steganography is the practice of hiding information
```

```
or'

Step 06 - j = 43
 token                          p_LM     xi      p_WM

messages                      0.4289  0.6481  0.1844
a                             0.1225  0.8668  0.1578
data                          0.0866  0.9805  0.4039
communication                 0.0451  0.9078  0.0595
communic                      0.0392  0.5786  0.0000
files                         0.0365  0.1152  0.0000
message                       0.0228  0.3551  0.0000
an                            0.0167  0.9419  0.0140
secret                        0.0140  0.9098  0.0006
other                         0.0129  0.8953  0.0001
 → Chosen token:  848  ('data')
 Current prompt so far → 'Steganography is the practice of hiding information or
data'

Step 07 - j = 44
 token                          p_LM     xi      p_WM

in                            0.4172  0.3880  0.0890
within                        0.3294  0.5037  0.1073
inside                        0.0450  0.9968  0.8022
by                            0.0322  0.5999  0.0000
into                          0.0249  0.7270  0.0000
,                             0.0169  0.3087  0.0000
.                             0.0123  0.8020  0.0000
(                             0.0118  0.0379  0.0000
so                            0.0101  0.6093  0.0000
from                          0.0075  0.9024  0.0000
 → Chosen token: 2768  ('inside')
 Current prompt so far → 'Steganography is the practice of hiding information or
data inside'

Step 08 - j = 45
 token                          p_LM     xi      p_WM

another                       0.3282  0.9122  0.5790
a                             0.1334  0.0519  0.0000
other                         0.1147  0.8349  0.1589
an                            0.1068  0.4628  0.0006
of                            0.0704  0.0595  0.0000
something                     0.0402  0.6814  0.0001
images                        0.0220  0.0028  0.0000
the                           0.0212  0.8157  0.0001
seem                          0.0178  0.4946  0.0000
```

```
innoc                        0.0124  0.3759  0.0000
 → Chosen token: 1790  ('another')
 Current prompt so far → 'Steganography is the practice of hiding information or
data inside another'

Step 09 - j = 46
 token                         p_LM     xi      p_WM

file                         0.2416  0.7254  0.6968
piece                        0.1735  0.0161  0.0000
object                       0.0766  0.8472  0.3017
message                      0.0727  0.4134  0.0000
data                         0.0664  0.3171  0.0000
form                         0.0549  0.2936  0.0000
medium                       0.0516  0.6087  0.0002
type                         0.0225  0.8214  0.0004
format                       0.0162  0.1777  0.0000
document                     0.0141  0.8934  0.0009
 → Chosen token:  934  ('file')
 Current prompt so far → 'Steganography is the practice of hiding information or
data inside another file'

Step 10 - j = 47
 token                         p_LM     xi      p_WM

.                            0.2530  0.9080  0.2473
,                            0.1886  0.4668  0.0064
or                           0.1714  0.9888  0.3393
in                           0.0369  0.9556  0.1057
format                       0.0366  0.5322  0.0000
(                            0.0353  0.9890  0.2649
so                           0.0343  0.5144  0.0000
such                         0.0311  0.0777  0.0000
that                         0.0296  0.2613  0.0000
and                          0.0273  0.3319  0.0000
 → Chosen token:  470  ('or')
 Current prompt so far → 'Steganography is the practice of hiding information or
data inside another file or'

Step 11 - j = 48
 token                         p_LM     xi      p_WM

data                         0.1141  0.0015  0.0000
document                     0.1038  0.3265  0.0000
object                       0.0913  0.5254  0.0007
message                      0.0892  0.2964  0.0000
image                        0.0811  0.2870  0.0000
file                         0.0456  0.1774  0.0000
```

```
a                           0.0432  0.6282  0.0000
stream                      0.0321  0.2360  0.0000
in                          0.0317  0.7812  0.0003
media                       0.0291  0.6347  0.0000
 → Chosen token: 7047  ('software')
 Current prompt so far → 'Steganography is the practice of hiding information or
data inside another file or software'

Step 12 - j = 49
 token                        p_LM     xi      p_WM

.                           0.3799  0.4944  0.0813
program                     0.0944  0.9283  0.2362
,                           0.0865  0.1092  0.0000
application                 0.0601  0.4776  0.0000
so                          0.0523  0.6839  0.0004
that                        0.0388  0.0989  0.0000
in                          0.0372  0.9299  0.0737
to                          0.0339  0.9705  0.2147
and                         0.0222  0.5515  0.0000
package                     0.0213  0.3590  0.0000
 → Chosen token: 1584  ('even')
 Current prompt so far → 'Steganography is the practice of hiding information or
data inside another file or software even'

Step 13 - j = 50
 token                        p_LM     xi      p_WM

when                        0.1881  0.4569  0.0737
though                      0.1371  0.6422  0.1878
without                     0.1288  0.5675  0.0583
if                          0.0987  0.8235  0.6642
in                          0.0696  0.2894  0.0000
the                         0.0389  0.7463  0.0026
.                           0.0281  0.4982  0.0000
with                        0.0190  0.3949  0.0000
before                      0.0168  0.0092  0.0000
while                       0.0163  0.5370  0.0000
 → Chosen token:  565  ('if')
 Current prompt so far → 'Steganography is the practice of hiding information or
data inside another file or software even if'

Step 14 - j = 51
 token                        p_LM     xi      p_WM

the                         0.3261  0.3363  0.0298
it                          0.3091  0.2754  0.0130
you                         0.0737  0.2147  0.0000
```

```
that                        0.0389  0.0529  0.0000
someone                     0.0293  0.0739  0.0000
this                        0.0267  0.0245  0.0000
they                        0.0203  0.9805  0.3194
there                       0.0174  0.8568  0.0001
its                         0.0124  0.9528  0.0170
a                           0.0097  0.9931  0.4121
 → Chosen token:  263  ('a')
 Current prompt so far → 'Steganography is the practice of hiding information or
data inside another file or software even if a'

Step 15 - j = 52
 token                        p_LM     xi      p_WM

third                       0.1508  0.5309  0.1289
file                        0.1174  0.5721  0.0736
ha                          0.1084  0.5504  0.0348
person                      0.1042  0.6590  0.1566
user                        0.0551  0.4802  0.0000
computer                    0.0287  0.2373  0.0000
mal                         0.0267  0.9193  0.3676
security                    0.0171  0.7249  0.0000
ste                         0.0168  0.0745  0.0000
program                     0.0152  0.0050  0.0000
 → Chosen token: 4439  ('mal')
 Current prompt so far → 'Steganography is the practice of hiding information or
data inside another file or software even if a mal'

Step 16 - j = 53
 token                        p_LM     xi      p_WM

icious                      0.7261  0.7160  0.7970
ware                        0.2686  0.6121  0.2030
war                         0.0016  0.7368  0.0000
ici                         0.0008  0.8824  0.0000
function                    0.0006  0.9768  0.0000
ice                         0.0004  0.1223  0.0000
formed                      0.0002  0.0271  0.0000
ic                          0.0002  0.7678  0.0000
-                           0.0002  0.1076  0.0000
p                           0.0001  0.7357  0.0000
 → Chosen token: 14803  ('icious')
 Current prompt so far → 'Steganography is the practice of hiding information or
data inside another file or software even if a malicious'

Step 17 - j = 54
 token                        p_LM     xi      p_WM
```

```
user                        0.1414  0.8372  0.3248
program                     0.0854  0.6036  0.0031
attack                      0.0773  0.3845  0.0000
person                      0.0738  0.0837  0.0000
software                    0.0709  0.7898  0.0410
party                       0.0538  0.9550  0.4845
actor                       0.0499  0.8823  0.0928
code                        0.0450  0.2514  0.0000
file                        0.0409  0.8820  0.0531
entity                      0.0386  0.5626  0.0000
 → Chosen token: 6263  ('party')
 Current prompt so far → 'Steganography is the practice of hiding information or
data inside another file or software even if a malicious party'

Step 18 - j = 55
 token                        p_LM     xi      p_WM

is                          0.1751  0.2536  0.0006
has                         0.1263  0.2013  0.0000
tries                       0.1072  0.9472  0.9346
can                         0.0507  0.6243  0.0001
knows                       0.0400  0.8242  0.0123
attempts                    0.0368  0.6832  0.0000
does                        0.0248  0.3643  0.0000
or                          0.0234  0.3882  0.0000
wants                       0.0216  0.5516  0.0000
would                       0.0205  0.0955  0.0000
 → Chosen token: 14335  ('tries')
 Current prompt so far → 'Steganography is the practice of hiding information or
data inside another file or software even if a malicious party tries'

Step 19 - j = 56
 token                        p_LM     xi      p_WM

to                          0.9826  0.2564  0.2145
.                           0.0016  0.1998  0.0000
hard                        0.0015  0.0973  0.0000
his                         0.0010  0.9888  0.0000
their                       0.0009  0.9999  0.7674
,                           0.0006  0.9978  0.0181
it                          0.0005  0.2482  0.0000
and                         0.0004  0.0998  0.0000
hack                        0.0004  0.9176  0.0000
its                         0.0004  0.2282  0.0000
 → Chosen token: 1009  ('their')
 Current prompt so far → 'Steganography is the practice of hiding information or
data inside another file or software even if a malicious party tries their'
---  END VERBOSE GENERATION  ---
```

```
Final partial generation:
 Steganography is the practice of hiding information or data inside another file
or software even if a malicious party tries their
```